

COUNTERFEIT IC DETECTION REPORT

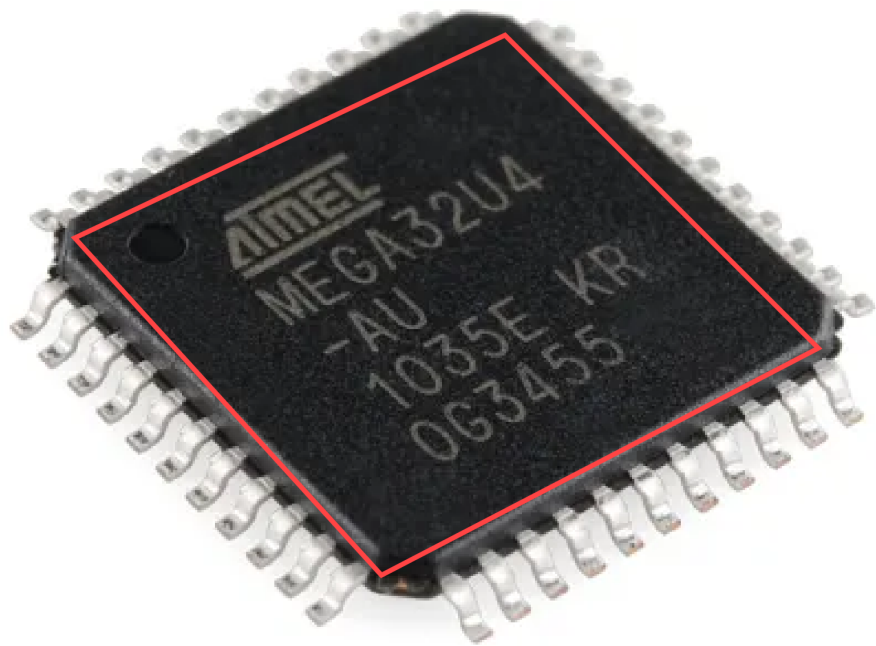
VERDICT	SUSPICIOUS - REQUIRES INSPECTION
AUTHENTICITY SCORE	68.8/100
TIMESTAMP	2025-12-07T22:11:04

IC INFORMATION

Part Number	UNKNOWN
Manufacturer	UNKNOWN
Package Type	UNKNOWN
Pin Count	0

INPUT IC IMAGES

Dimension Analysis - Detected IC Body



IC Body Detected
219x203 px
AR: 1.077

EXECUTIVE SUMMARY

Overall Assessment: UNKNOWN

Text Quality Analysis: X POOR (Score: 0/100)

Markings show poor quality with visible erosion, inconsistent font weight, or irregular spacing. Possible signs of remarking or tampering detected.

Pin Count Verification: X MISMATCH

Pin count mismatch detected. Expected 0 pins according to datasheet, but actual count differs. This is a significant indicator of potential counterfeiting.

Package Type Verification: X MISMATCH

Package type mismatch detected. Expected UNKNOWN but physical characteristics differ from datasheet. This indicates potential counterfeiting.

Key Observations:

- 4
- 2
- 9
-
- Y

Dimensional Analysis:

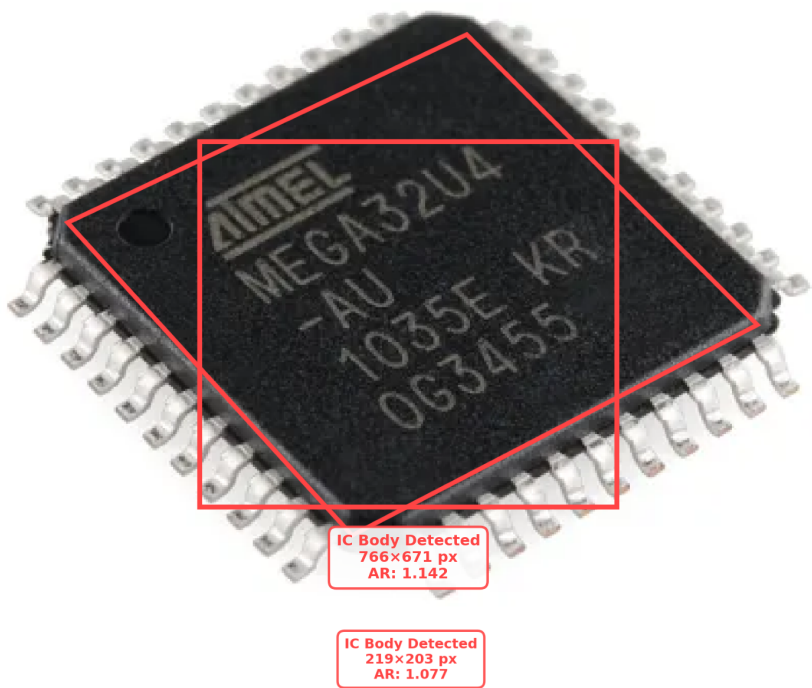
Measured aspect ratio: 1.142. No ground truth available for comparison.

Final Reasoning: Dimensional analysis: 100.0/100; Visual assessment: unknown

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body

Dimension Analysis - Detected IC Body



Parameter	Value
Measured Aspect Ratio	1.142
Body Width (px)	766
Body Height (px)	671
Dimension Source	no_ground_truth
Dimension Score	100.0/100
Verdict	MATCH

DATASHEET OUTLINE DIMENSION PAGE

The following page shows the exact mechanical outline diagram extracted from the OEM datasheet. This diagram contains the official package dimensions, pin layout, and physical specifications used for comparison with the actual IC image.

Designing DC/DC converters based on SEPIC topology

By Jeff Falin
Senior Applications Engineer

Introduction

The single-ended primary-inductance converter (SEPIC) is a DC/DC-converter topology that provides a positive regulated output voltage from an input voltage that varies from above to below the output voltage. This type of conversion is handy when the designer uses voltages (e.g., 12 V) from an unregulated input power supply such as a low-cost wall wart. Unfortunately, the SEPIC topology is difficult to understand and requires two inductors, making the power-supply footprint quite large. Recently, several inductor manufacturers began selling off-the-shelf coupled inductors in a single package at a cost only slightly higher than that of the comparable single inductor. The coupled inductor not only provides a smaller footprint but also, to get the same inductor ripple current, requires only half the inductance required for a SEPIC with two separate inductors. This article explains how to design a SEPIC converter with a coupled inductor.

Basic operation

Figure 1 shows a simple circuit diagram of a SEPIC converter, consisting of an input capacitor, C_{IN} ; an output capacitor, C_{OUT} ; coupled inductors $L1a$ and $L1b$; an AC

coupling capacitor, C_P ; a power FET, $Q1$; and a diode, $D1$. Figure 2 shows the SEPIC operating in continuous conduction mode (CCM). $Q1$ is on in the top circuit and off in the bottom circuit.

To understand the voltages at the various circuit nodes, it is important to analyze the circuit at DC when $Q1$ is off and not switching. During steady-state CCM, pulse-width-modulation (PWM) operation, and neglecting ripple voltage,

Figure 1. Simple circuit diagram of SEPIC converter

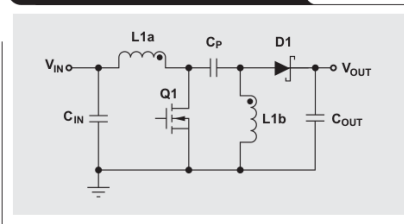
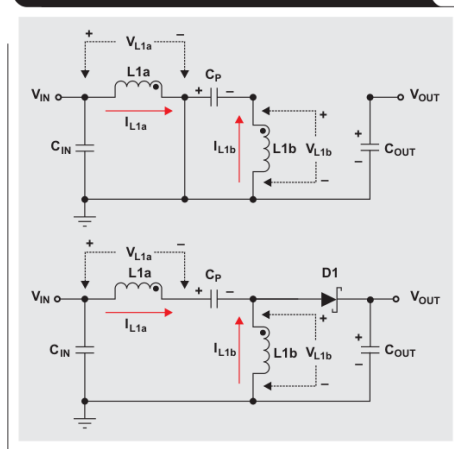


Figure 2. SEPIC during CCM operation when Q1 is on (top) and off (bottom)



EXTRACTED DIMENSIONS FROM DIAGRAM

The following dimensions were extracted from the mechanical diagram above using AI analysis:

Parameter	Value
Package Type	UNKNOWN

OEM DATASHEET REFERENCE

Source File: UNKNOWN_tavily.pdf
Location: /Users/shnjnmkkr/Desktop/SIH
again/counterfeit_IC/data/personal_api_results/datasheets/UNKNOWN_tavily.pdf

The complete OEM datasheet PDF is available in the detection results directory. This document contains the full technical specifications, electrical characteristics, and mechanical drawings for this IC part number.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 0/100

Status: X POOR

The IC markings show significant quality issues that raise concerns:

- Inconsistent font weight, spacing, or alignment
- Visible erosion, fading, or irregular printing
- Possible signs of remarking or surface tampering
- Markings do not match OEM quality standards
- This is a strong indicator of potential counterfeiting

2. Pin Configuration Analysis

Pin Count: X MISMATCH

Pin count discrepancy detected. Expected 0 pins according to datasheet, but actual count differs. This is a significant red flag indicating potential counterfeiting or incorrect part identification.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: X MISMATCH

Package type discrepancy detected. Expected UNKNOWN but physical characteristics differ from datasheet. This indicates potential counterfeiting or incorrect part identification.

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: UNKNOWN

Visual assessment completed. Review detailed analysis above for specific findings.

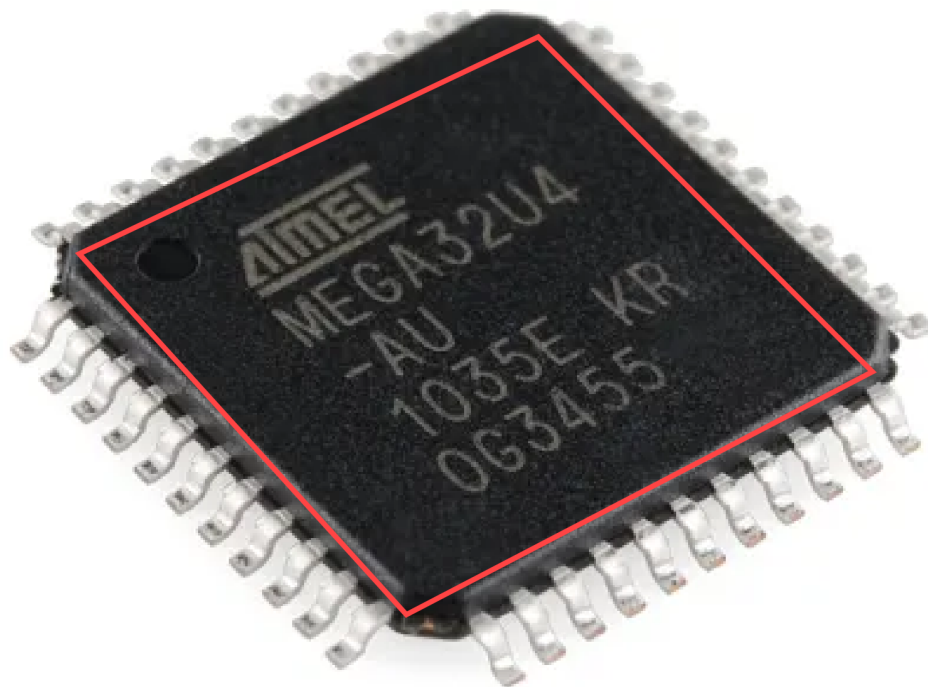
ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 100.0/100; Visual assessment: unknown

DETECTED ANOMALIES (0)

No anomalies detected. IC appears authentic.

□ NO ANOMALIES DETECTED Dimension Analysis - Detected IC Body



IC Body Detected
219×203 px
AR: 1.077